Simulink Design Verifier Report

FeedbackControl

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Simulink Design Verifier Report: FeedbackControl

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Chapter 1. Summary

Analysis Information.

Model:	FeedbackControl
Release:	R2026a Prerelease
Checksum:	2003161129 2501949746 1271412842 2870760681
Mode:	Test generation
Model Representation:	Built on 15-May-2026 08:41:58
Test Generation Target:	Model
Status:	Completed normally
PreProcessing Time:	7.9688s
Analysis Time:	16s

Objectives Status.

Number of Objectives:	**43**	
Objectives Satisfied:	39	(91%)
Objectives Unsatisfiable:	2	(5%)
Objectives Undecided Due to Runtime Error:	2	(4%)

Chapter 2. Analysis Information

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Model Information

File:	FeedbackControl
Version:	8.7
Time Stamp:	Fri May 15 08:39:33 2026
Author:	The MathWorks, Inc.

Analysis Options

Mode:	TestGeneration
Rebuild Model Representation:	IfChangeIsDetected
Test Generation Target:	Model
Test Suite Optimization:	Auto
Maximum Testcase Steps:	10000time steps
Test Conditions:	UseLocalSettings
Test Objectives:	UseLocalSettings
Model Coverage Objectives:	MCDC
Add tests for the missing coverage:	off
Include Relational Boundary Objectives:	off
Maximum Analysis Time:	180s
Block Replacement:	off
Parameters Analysis:	off
Include expected output values:	off
Randomize data that do not affect the outcome:	off
Additional analysis to reduce instances of rational approximation:	on
Save Harness:	off
Save Report:	on

User Artifacts

Coverage Data:	n/a
Test Data:	n/a

Unsupported Blocks

The following blocks are not supported by Simulink Design Verifier. They were abstracted during the analysis. This can lead Simulink Design Verifier to produce only partial results for parts of the model that depends on the output values of these blocks.

Block	**Type**
GuidanceCompute/MATLAB Function	chart
HealthMonitor/FaultDetector	chart

Approximations

Simulink Design Verifier performed the following approximations during analysis. These can impact the precision of the results generated by Simulink Design Verifier. Please see the product documentation for further details.

#	**Type**	**Description**
1	Rational approximation	The model includes floating-point arithmetic. Simulink Design Verifier approximates floating-point arithmetic with rational number arithmetic. Specifying minimum and maximum values that mimic environmental constraints on root-level Inport blocks may reduce instances of rational approximation.

Chapter 3. Test Objectives Status

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Objectives Satisfied

Simulink Design Verifier generated test cases that exercise these test objectives.

#	Type	Model Item	Description	Analysis Time (sec)	Test Case
1	Decision	GuidanceCompute/MATLAB Function	function y = vector_norm(u) **executed**	8	1 [13]
2	Decision	Chart "GuidanceCompute/Chart"	Substate executed **State "NearWP"**	12	2 [15]
3	Decision	Chart "GuidanceCompute/Chart"	Substate executed **State "NotNearWP"**	8	1 [13]
4	Decision	Chart "GuidanceCompute/Chart"	Substate executed **State "WPachieved"**	13	3 [15]
5	Decision	Multiport Switch	integer input value = **0 (output is from input port 0)**	8	1 [13]
6	Decision	Multiport Switch	integer input value = **1 (output is from input port 1)**	8	1 [13]
7	Decision	Multiport Switch	integer input value = **2 (output is from input port 2)**	8	1 [13]
8	Decision	Multiport Switch	integer input value = ***,3 (output is from input port 3)**	8	1 [13]
9	Decision	HealthMonitor/FaultDetector	function [ThrustAnomaly, ModeAnomaly, PositionAnomaly, AnyFault] = detectFaults(Thrust, Mode, RelPos) **executed**	8	1 [13]
10	Condition	HealthMonitor/FaultDetector	Thrust < 0 **true**	8	1 [13]
11	Condition	HealthMonitor/FaultDetector	Thrust < 0 **false**	8	1 [13]

#	Type	Model Item	Description	Analysis Time (sec)	Test Case
12	Condition	HealthMonitor/FaultDetector	Thrust > 2.0 **true**	8	1 [13]
13	Condition	HealthMonitor/FaultDetector	Thrust > 2.0 **false**	8	1 [13]
15	Condition	HealthMonitor/FaultDetector	any(isnan(RelPos)) **false**	8	1 [13]
16	Condition	HealthMonitor/FaultDetector	norm(double(RelPos)) > 10.0 **true**	8	1 [13]
17	Condition	HealthMonitor/FaultDetector	norm(double(RelPos)) > 10.0 **false**	8	1 [13]
18	Condition	HealthMonitor/FaultDetector	ThrustAnomaly **true**	8	1 [13]
19	Condition	HealthMonitor/FaultDetector	ThrustAnomaly **false**	8	1 [13]
21	Condition	HealthMonitor/FaultDetector	ModeAnomaly **false**	8	1 [13]
22	Condition	HealthMonitor/FaultDetector	PositionAnomaly **true**	8	1 [13]
23	Condition	HealthMonitor/FaultDetector	PositionAnomaly **false**	8	1 [13]
24	MCDC	HealthMonitor/FaultDetector	Thrust < 0 || Thrust > 2.0 with Thrust < 0 **true**	8	1 [13]
25	MCDC	HealthMonitor/FaultDetector	Thrust < 0 || Thrust > 2.0 with Thrust < 0 **false**	8	1 [13]
26	MCDC	HealthMonitor/FaultDetector	Thrust < 0 || Thrust > 2.0 with Thrust > 2.0 **true**	8	1 [13]
27	MCDC	HealthMonitor/FaultDetector	Thrust < 0 || Thrust > 2.0 with Thrust > 2.0 **false**	8	1 [13]
29	MCDC	HealthMonitor/FaultDetector	any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with any(isnan(RelPos)) **false**	8	1 [13]
30	MCDC	HealthMonitor/FaultDetector	any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with norm(double(RelPos)) > 10.0 **true**	8	1 [13]
31	MCDC	HealthMonitor/FaultDetector	any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with norm(double(RelPos)) > 10.0 **false**	8	1 [13]
32	MCDC	HealthMonitor/FaultDetector	ThrustAnomaly || ModeAnomaly || PositionAno-	8	1 [13]

#	Type	Model Item	Description	Analysis Time (sec)	Test Case
			maly with ThrustAnomaly **true**		
33	MCDC	HealthMonitor/FaultDetector	ThrustAnomaly || ModeAnomaly || PositionAnomaly with ThrustAnomaly **false**	8	1 [13]
35	MCDC	HealthMonitor/FaultDetector	ThrustAnomaly || ModeAnomaly || PositionAnomaly with ModeAnomaly **false**	8	1 [13]
36	MCDC	HealthMonitor/FaultDetector	ThrustAnomaly || ModeAnomaly || PositionAnomaly with PositionAnomaly **true**	8	1 [13]
37	MCDC	HealthMonitor/FaultDetector	ThrustAnomaly || ModeAnomaly || PositionAnomaly with PositionAnomaly **false**	8	1 [13]
38	Decision	Transition "[WPerr > 0.1]" from "NearWP" to "NotNearWP"	expression "WPerr > 0.1" **true**	12	2 [15]
39	Decision	Transition "[WPerr > 0.1]" from "NearWP" to "NotNearWP"	expression "WPerr > 0.1" **false**	13	3 [15]
40	Decision	Transition "[count >= 0.75]" from "NearWP" to "WPachieved"	expression "count >= 0.75" **true**	13	3 [15]
41	Decision	Transition "[count >= 0.75]" from "NearWP" to "WPachieved"	expression "count >= 0.75" **false**	13	3 [15]
42	Decision	Transition "[WPerr <= 0.1]" from "NotNearWP" to "NearWP"	expression "WPerr <= 0.1" **true**	12	2 [15]
43	Decision	Transition "[WPerr <= 0.1]" from "NotNearWP" to "NearWP"	expression "WPerr <= 0.1" **false**	8	1 [13]

Objectives Unsatisfiable

Simulink Design Verifier proved these test objectives to be unreachable by any test case. This often indicates the presence of dead logic in the model. This can be a side effect of parameter configurations or minimum and maximum constraints specified on inputs. In Test Generation, this can also be a result of constraints resulting from Test Condition blocks.

#	Type	Model Item	Description	Analysis Time (sec)
14	Condition	HealthMonitor/FaultDetector	any(isnan(RelPos)) **true**	9
28	MCDC	HealthMonitor/FaultDetector	any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with any(isnan(RelPos)) **true**	9

Objectives Undecided Due to Runtime Error

Simulink Design Verifier was not able to decide these objectives due to runtime errors that occurred during simulation of the test cases.

#	Type	Model Item	Description	Analysis Time (sec)	Test Case
20	Condition	HealthMonitor/FaultDetector	ModeAnomaly **true**	15	4 [16]
34	MCDC	HealthMonitor/FaultDetector	ThrustAnomaly || ModeAnomaly || PositionAnomaly with ModeAnomaly **true**	15	4 [16]

Chapter 4. Model Items

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Transition "[count >= 0.75]" from "NearWP" to "WPachieved" 11
Transition "[WPerr <= 0.1]" from "NotNearWP" to "NearWP" 12

This section presents, for each object in the model defining coverage objectives, the list of objectives and their individual status at the end of the analysis. It should match the coverage report obtained from running the generated test suite on the model, either from the harness model or by using the sldvruntest command.

GuidanceCompute/MATLAB Function

#:	Type	Description	Status	Test Case
1	Decision	function y = vector_norm(u) executed	Satisfied	1 [13]

Chart "GuidanceCompute/Chart"

#:	Type	Description	Status	Test Case
2	Decision	Substate executed State "NearWP"	Satisfied	2 [15]
3	Decision	Substate executed State "NotNearWP"	Satisfied	1 [13]
4	Decision	Substate executed State "WPachieved"	Satisfied	3 [15]

Multiport Switch

#:	Type	Description	Status	Test Case
5	Decision	integer input value = 0 (output is from input port 0)	Satisfied	1 [13]
6	Decision	integer input value = 1 (output is from input port 1)	Satisfied	1 [13]

#:	Type	Description	Status	Test Case
7	Decision	integer input value = 2 (output is from input port 2)	Satis-fied	1 [13]
8	Decision	integer input value = *,3 (output is from in-put port 3)	Satis-fied	1 [13]

HealthMonitor/FaultDetector

#:	Type	Description	Status	Test Case
9	Decision	function [ThrustAno-maly, ModeAnoma-ly, PositionAnomaly, AnyFault] = detect-Faults(Thrust, Mode, RelPos) executed	Satis-fied	1 [13]
10	Condition	Thrust < 0 true	Satis-fied	1 [13]
11	Condition	Thrust < 0 false	Satis-fied	1 [13]
12	Condition	Thrust > 2.0 true	Satis-fied	1 [13]
13	Condition	Thrust > 2.0 false	Satis-fied	1 [13]
14	Condition	any(isnan(RelPos)) true	Unsat-isfia-ble	n/a
15	Condition	any(isnan(RelPos)) false	Satis-fied	1 [13]
16	Condition	norm(double(RelPos)) > 10.0 true	Satis-fied	1 [13]
17	Condition	norm(double(RelPos)) > 10.0 false	Satis-fied	1 [13]
18	Condition	ThrustAnomaly true	Satis-fied	1 [13]
19	Condition	ThrustAnomaly false	Satis-fied	1 [13]
20	Condition	ModeAnomaly true	Unde-cided due to run-time error	4 [16]

#:	Type	Description	Status	Test Case
21	Condition	ModeAnomaly false	Satisfied	1 [13]
22	Condition	PositionAnomaly true	Satisfied	1 [13]
23	Condition	PositionAnomaly false	Satisfied	1 [13]
24	MCDC	Thrust < 0 || Thrust > 2.0 with Thrust < 0 true	Satisfied	1 [13]
25	MCDC	Thrust < 0 || Thrust > 2.0 with Thrust < 0 false	Satisfied	1 [13]
26	MCDC	Thrust < 0 || Thrust > 2.0 with Thrust > 2.0 true	Satisfied	1 [13]
27	MCDC	Thrust < 0 || Thrust > 2.0 with Thrust > 2.0 false	Satisfied	1 [13]
28	MCDC	any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with any(isnan(RelPos)) true	Unsatisfiable	n/a
29	MCDC	any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with any(isnan(RelPos)) false	Satisfied	1 [13]
30	MCDC	any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with norm(double(RelPos)) > 10.0 true	Satisfied	1 [13]
31	MCDC	any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with norm(double(RelPos)) > 10.0 false	Satisfied	1 [13]
32	MCDC	ThrustAnomaly || ModeAnomaly || PositionAnomaly with ThrustAnomaly true	Satisfied	1 [13]
33	MCDC	ThrustAnomaly || ModeAnomaly || PositionAnomaly with ThrustAnomaly false	Satisfied	1 [13]
34	MCDC	ThrustAnomaly || ModeAnomaly || Po-	Undecided due to	4 [16]

#:	Type	Description	Status	Test Case
		sitionAnomaly with ModeAnomaly true	run-time error	
35	MCDC	ThrustAnomaly || ModeAnomaly || PositionAnomaly with ModeAnomaly false	Satisfied	1 [13]
36	MCDC	ThrustAnomaly || ModeAnomaly || PositionAnomaly with PositionAnomaly true	Satisfied	1 [13]
37	MCDC	ThrustAnomaly || ModeAnomaly || PositionAnomaly with PositionAnomaly false	Satisfied	1 [13]

Transition "[WPerr > 0.1]" from "NearWP" to "NotNearWP"

#:	Type	Description	Status	Test Case
38	Decision	expression "WPerr > 0.1" true	Satisfied	2 [15]
39	Decision	expression "WPerr > 0.1" false	Satisfied	3 [15]

Transition "[count >= 0.75]" from "NearWP" to "WPachieved"

#:	Type	Description	Status	Test Case
40	Decision	expression "count >= 0.75" true	Satisfied	3 [15]
41	Decision	expression "count >= 0.75" false	Satisfied	3 [15]

Transition "[WPerr <= 0.1]" from "NotNearWP" to "NearWP"

#:	Type	Description	Status	Test Case
42	Decision	expression "WPerr <= 0.1" true	Satisfied	2 [15]
43	Decision	expression "WPerr <= 0.1" false	Satisfied	1 [13]

Chapter 5. Test Cases

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This section contains detailed information about each generated test case. Some input signals are unused. Input data will not be reported for them.

Test Case 1

Summary.

Length:	0.1 second (11 sample periods)
Objectives Satisfied:	32

Objectives.

Step	Time	Model Item	Objectives
1	0	GuidanceCompute/MATLAB Function Multiport Switch HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector	1. function y = vector_norm(u) executed [4] 6. integer input value = 1 (output is from input port 1) [4] 9. function [ThrustAnomaly, ModeAnomaly, PositionAnomaly, AnyFault] = detectFaults(Thrust, Mode, RelPos) executed [4] 11. Thrust < 0 false [4] 12. Thrust > 2.0 true [5] 15. any(isnan(RelPos)) false [5] 16. norm(double(RelPos)) > 10.0 true [5] 18. ThrustAnomaly true [5] 26. Thrust < 0 || Thrust > 2.0 with Thrust > 2.0 true [5] 30. any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with norm(double(RelPos)) > 10.0 true [5] 32. ThrustAnomaly || ModeAnomaly || PositionAnomaly with ThrustAnomaly true [5]
2	0.01	Chart "GuidanceCompute/Chart" Transition "[WPerr <= 0.1]" from "NotNearWP" to "NearWP"	3. Substate executed State "NotNearWP" [4] 43. expression "WPerr <= 0.1" false [6]

Step	Time	Model Item	Objectives
4	0.03	Multiport Switch HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector	5. integer input value = 0 (output is from input port 0) [4] 13. Thrust > 2.0 false [5] 19. ThrustAnomaly false [5] 21. ModeAnomaly false [5] 22. PositionAnomaly true [5] 25. Thrust < 0 || Thrust > 2.0 with Thrust < 0 false [5] 27. Thrust < 0 || Thrust > 2.0 with Thrust > 2.0 false [5] 36. ThrustAnomaly || ModeAnomaly || PositionAnomaly with PositionAnomaly true [6]
5	0.04	Multiport Switch	7. integer input value = 2 (output is from input port 2) [4]
6	0.05	Multiport Switch	8. integer input value = *,3 (output is from input port 3) [4]
10	0.09	HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector	10. Thrust < 0 true [4] 17. norm(double(RelPos)) > 10.0 false [5] 24. Thrust < 0 || Thrust > 2.0 with Thrust < 0 true [5] 29. any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with any(isnan(RelPos)) false [5] 31. any(isnan(RelPos)) || norm(double(RelPos)) > 10.0 with norm(double(RelPos)) > 10.0 false [5]
11	0.1	HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector HealthMonitor/FaultDetector	23. PositionAnomaly false [5] 33. ThrustAnomaly || ModeAnomaly || PositionAnomaly with ThrustAnomaly false [6] 35. ThrustAnomaly || ModeAnomaly || PositionAnomaly with ModeAnomaly false [6] 37. ThrustAnomaly || ModeAnomaly || PositionAnomaly with PositionAnomaly false [6]

Generated Input Data.

Time	**0-0.02**	**0.03**	**0.04**	**0.05-0.06**	**0.07**	**0.08**	**0.09-0.1**
Step	**1-3**	**4**	**5**	**6-7**	**8**	**9**	**10-11**
Mode	1	0	2	3	2	1	1
UAV-Pos.WorldPosition	[-8.5493 7.5802 6.1676]	[-8.5493 7.5802 6.1676]	[-8.5493 7.5802 6.1676]	[-8.5493 7.5802 6.1676]	[3.9681 -9.9985 4.6623]	[-7.9535 9.4032 7.4182]	[3.4497 7.1965 4.0577]

Test Case 2

Summary.

Length:	0.02 second (3 sample periods)
Objectives Satisfied:	3

Objectives.

Step	Time	Model Item	Objectives
2	0.01	Transition "[WPerr <= 0.1]" from "NotNearWP" to "NearWP"	42. expression "WPerr <= 0.1" true [6]
3	0.02	Chart "GuidanceCompute/Chart" Transition "[WPerr > 0.1]" from "NearWP" to "NotNearWP"	2. Substate executed State "NearWP" [4] 38. expression "WPerr > 0.1" true [6]

Generated Input Data.

Time	0	0.01	0.02
Step	**1**	**2**	**3**
Mode	0	0	1
UAVPos.WorldPosition	[8.495e-91 1501 -285.75]	[0.007473 8.6917e-311 -1]	[-8.5493 7.5802 6.1676]

Test Case 3

Summary.

Length:	0.78 second (79 sample periods)
Objectives Satisfied:	4

Objectives.

Step	Time	Model Item	Objectives
3	0.02	Transition "[WPerr > 0.1]" from "NearWP" to "NotNearWP" Transition "[count >= 0.75]" from "NearWP" to "WPachieved"	39. expression "WPerr > 0.1" false [6] 41. expression "count >= 0.75" false [6]
78	0.77	Transition "[count >= 0.75]" from "NearWP" to "WPachieved"	40. expression "count >= 0.75" true [6]
79	0.78	Chart "GuidanceCompute/Chart"	4. Substate executed State "WPachieved" [4]

Generated Input Data.

Time	0	0.01-0.78
Step	**1**	**2-79**
Mode	0	3
UAVPos.WorldPosition	[8.495e-91 1501 -285.75]	[0.007473 0 -1]

Test Case 4

Summary.

Length:	0 second (1 sample period)
Objectives Satisfied:	2
Runtime error description:	For block 'FeedbackControl/Multiport Switch', control port value '128' is not in between '0' and '3'. To suppress this message and use the default port, you can change the settings of 'Diagnostic for default case' to 'None'

Objectives.

Step	Time	Model Item	Objectives
1	0	HealthMonitor/FaultDetector HealthMonitor/FaultDetector	20. ModeAnomaly true [7] 34. ThrustAnomaly || ModeAnomaly || PositionAnomaly with ModeAnomaly true [7]

Generated Input Data.

Time	0
Step	**1**
Mode	128
UAVPos.WorldPosition	[2.743062034396844e+303 4.185580496821357e+298 -0.99219]